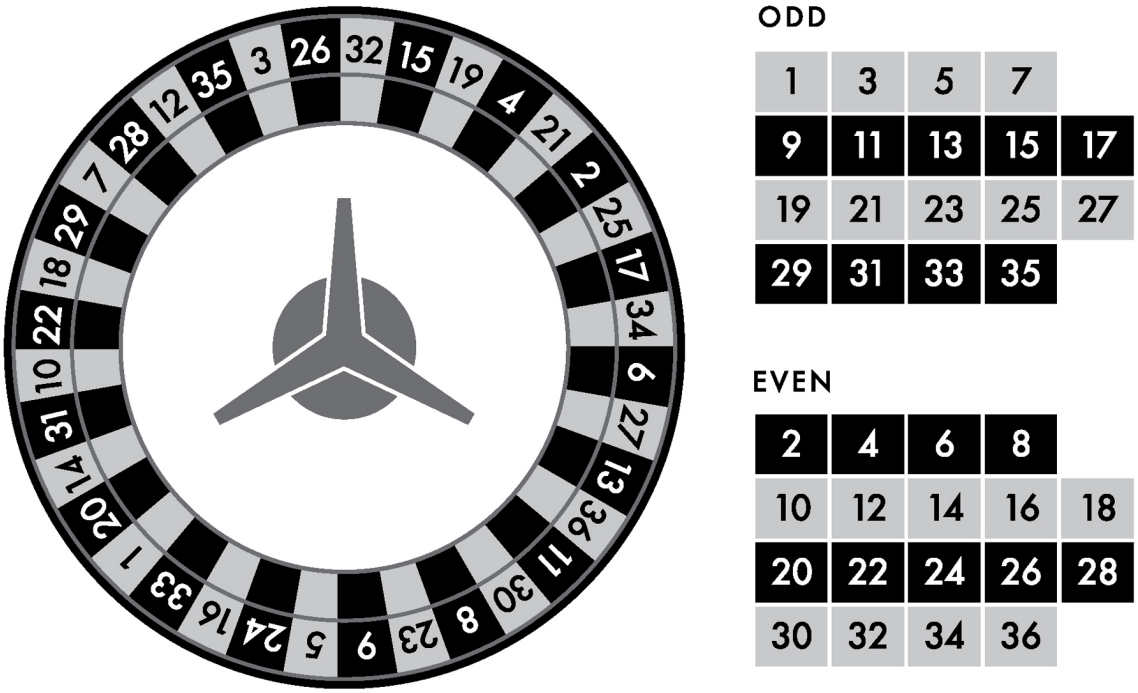


Figure 7.2. A fair roulette wheel. The numbers are split evenly between high-low, black-red, and even-odd, with no 0 or 00 slots.

But if the roulette wheel is contextual, all that goes out the window. For a contextual roulette wheel, the answer to the question, “Is the ball on a red number?” depends on what else you ask at the same time. Say you ask whether the ball is on a red number and whether it’s on an even number after a particular spin of the wheel. As it turns out, the answer to both your questions is yes. But if you had asked different questions after the same spin of the wheel—if you had instead asked whether the ball is on a red number and whether it’s on a high number—both answers would have been no. Somehow, the answer to the question, “Is the ball on a red number?” is actually affected by which other question you ask! This is contextuality: the answer to a question depends on its context of surrounding questions asked at the same time. In demolishing the no-hidden-variables proofs, Bell also demonstrated that quantum physics describes a contextual world.

At first blush, the fact that quantum physics is contextual seems to support the Copenhagen interpretation, or something like it. If the answers to questions depend on the other questions asked along with them, doesn’t